CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Sixth Semester of B. Tech. (EC) Examination

May 2014

EC311 VLSI Technology & Design

Date: 13/05/2014 **Time: 10:00 am to 01:00 pm** **Maximum Marks: 70**

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

Q - 1 (a) Fill in the blanks: [02]

(i) The Y Chart consists of three domains of representation, namely (1) __________, (2) ____________, and (3) ____________.

(ii) The value of the gate-to-source voltage V_{GS} needed to cause surface inversion is called the ________.

(iii) A MOS transistor which has no conducting channel region at zero gate bias is called a/an ___________ MOSFET.

(b) List down the designable parameters in a depletion load inverter circuit. [02]

(c) Discuss LOCOS process. [03]

Q - 2 (a) Explain MOS capacitor under external bias with energy band diagram. [08]

(b) Derive the threshold voltage (V_{th}) for CMOS inverter and prove $(W/L)_p \cong 2.5\ (W/L)_n$ for symmetric CMOS inverter. [06]

OR

Q - 2 (a) Explain Delay Time and calculate expression for T_{PHL} using Differential Equation method. [06]

(b) For the Resistor load inverter V_{DD}=3 V. Find the value of R_L such that V_0=0.1V. Assume transistor parameters: $K_n' = 60\ \mu A/v^2$, W/L = 5 and $V_{T0,N}$ = 0.5V. [04]

(c) Explain the types of photo resist used in fabrication process. [04]

Q - 3 (a) Draw and Explain CMOS D-latch and Edge Triggered Master Slave D Flip-flop with its Timing Diagram. [08]

(b) Consider the following MOSFET parameters. [06]

Substrate doping $N_D = 10^{15}$ cm^{-3}, Polysilicon gate doping density $N_D = 10^{20}$ cm^{-3}, Gate oxide thickness t_{ox}=650 A^0, and oxide interface charge density $N_{0X} = 2*10^{10}$ cm^{-2}. Use $\varepsilon_{ox} = 3.97\ \varepsilon_0$, $\varepsilon_{si} = 11.7\varepsilon_0$, where $\varepsilon_0 = 8.854*10^{-14}$ F/cm.

1) Calculate the threshold voltage V_{T0} for $V_{SB} = 0$
2) Determine the type and the amount of channel ion implantation which are necessary to achieve a threshold voltage of V_{T0} = -2V.

OR

Q - 3 (a) Discuss basic steps for fabrication and briefly explain p-well CMOS fabrication process. [07]

(b) Realized the following Boolean function using CMOS logic circuit. Assume (W/L)p=15 for all pMOS transistors and (W/L)n= 10 for all nMOS transistors then find out (W/L)n.equivalent and (W/L)p.equivalent for the circuit. [05]

Z= [A(D+E) +BC]'

(c) What is meant by ratioed logic? [02]

SECTION – II

Q - 4 (a) Realize the following Boolean function using CMOS transmission gates: [02]

(i) Two input XOR gate.

(ii) F= AB+A'C'+AB'C

(b) Draw the circuit diagram of CMOS SR latch circuit based on NAND2 gates. [02]

(c) What is the minimum power supply required on CMOS inverter so that it can operate properly? [03]

Q - 5 (a) Consider the depletion-load nMOS NOR2 gate with following parameters: $\mu_n C_{ox}$=25 $\mu A/V^2$, $V_{T0,driver}$ = 1 V, $V_{T0,load}$=-3 V, γ=0.4 $V^{1/2}$, and $|2\phi_F|$=0.6V. The transistor dimensions are given as $(W/L)_A$=2 , $(W/L)_B$= 4, and $(W/L)_{load}$ = 1/3. The power supply voltage is V_{DD}=5 V. Neglect substrate bias effect of load transistor. [06]

Calculate the output voltage levels for all four valid input voltage combinations.

(b) Explain the Basic principle of Pass transistor for logic "1" transfer and find out t_{charge} equation for the same. [06]

(c) Does the inverter with a lower V_{OL} always have the shorter high-to-low switching time? [02]

OR

Q – 5 (a) Discuss the various techniques available for on chip clock generation and distribution for successful high-speed VLSI design. [07]

(b) Explain Dynamic CMOS logic (Precharge – Evaluate logic) in detail. Also illustrate the cascading problem in dynamic CMOS logic. [07]

Q – 6 (a) Explain Ad-hoc Testable Design Techniques in Detail. [06]

(b) What is Noise margin for digital circuits? [05]

Calculate the V_{OH}, V_{OL} and V_{th} for CMOS inverter with the following parameters:

nMOS: $V_{T0,n} = 0.6V$ $\mu nCox = 60\ \mu A/V^2$ $(W/L)n = 8$

pMOS : $V_{T0,p} = -0.7V$ $\mu pCox = 25\ \mu A/V^2$ $(W/L)p = 12$

V_{DD}=3.3 V

(c) Define the following terms with reference to VLSI design process: [03]

Regularity, Modularity, Locality

OR

Q - 6 (a) Compare the two technology scaling methods, namely, (i) the constant electric-field scaling and (ii) the constant power-supply voltage scaling. In particular, show analytically by using equations how the delay time ,power dissipation and power density are affected in terms of the scaling factor, S. [06]

(b) Consider a CMOS ring oscillator consisting of an odd number (n) of identical inverters connected in a ring configuration. Derive an expression for the intrinsic delay (τ_P) in terms of the number of stages n. [04]

(c) Explain Latch-up Problem in CMOS Inverter, Mention Causes of Latch up problem. [04]

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